

CSE-4632

Digital Signal Processing Lab

Lab 2

Lab Group: **2**

Thursday, 4th June, 2026

Task 1:

Background.

A continuous-time signal

$$x(t) = A\sin(2\pi f_0 t)$$

is sampled at rate f_s to obtain the discrete-time sequence

$$x[n] = x(nT_s)$$

where $T_s = 1/f_s$.

Tasks:

1. Generate the analogue signal

$$x(t) = \sin(2\pi \cdot 5t)$$

over the interval $t \in [0, 1]$.

2. Sample the signal at:

$$f_s \in \{8,\ 5,\ 3\}\ \text{Hz}$$

3. Reconstruct the sampled signal using Zero-Order Hold (ZOH).

4. Plot:

- original signal,
- sampled points,
- reconstructed signal.

Task 2:

$$x[n] = \{3,\ 1,\ -2,\ 4,\ 2\}$$

Tasks:

1. Decompose the signal into shifted impulses.

2. Plot each impulse component separately.

3. Sum all components.

4. Verify the reconstructed signal.

Task 3:

$$x_e[n] = \frac{x[n] + x[-n]}{2}$$

$$x_o[n] = \frac{x[n] - x[-n]}{2}$$

Tasks:

1. Define:
$$x[n] = \{1, 2, 3, 4, 5\}$$
2. Compute:
$$x_e[n]$$
and
$$x_o[n]$$
3. Verify:
$$x[n] = x_e[n] + x_o[n]$$
4. Plot all signals.

Task 4:

Tasks:

1. Generate:
$$x[n] = (0.7)^n u[n]$$
2. Compute:
$$y[n] = x[-n]$$
3. Plot original and folded signals.

Task 5:

Tasks:

1. Define:
$$x[n] = \{1, 2, 3, 4, 3, 2, 1\}$$
2. Perform:
   - folding,
   - shifting,
   - combined transformation.
3. Verify:
$$y[n] = x[-n + 2]$$
4. Plot intermediate and final signals.

Task 6:

$$x[n] = 2\delta[n + 2] - \delta[n] + 3\delta[n - 1] + \delta[n - 3]$$

Tasks:

1. Construct the signal using impulses.
2. Compute:
   - $x[n - 2]$
   - $x[-n]$
   - $x[-n + 2]$
   - $r[n]x[n]$
3. Perform even–odd decomposition of:
$$x[-n + 2]$$
4. Plot all resulting signals.